Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$ (1) K_c = 0.119 (1) $dm^6\ mol^{-2}$ (1) student is correct since $K_c < 1$ / very small (1)		3	1	4	1 1	
		(ii)		student incorrect since equilibrium shifts to left (1) system opposes change by taking in heat, favouring endothermic direction (1)	1		1	2		
	(b)			$n(NH_3)$ = 58720 (1) n(DAP) = 22020 (1) maximum mass DAP = 2909 (1)		3		3	1 1	
	(c)			pV = nRT (1) $V = \frac{2.54 \times 10^{-3} \times 8.31 \times 393}{101000}$ (1) $V = 8.21 \times 10^{-5}\ m^3$ (1) $V = 82.1\ cm^3$ (1) credit alternative method using molar volume	1	3		4	1 1 1	
				Question 10 total	**2**	**9**	**2**	**13**	**7**	**0**